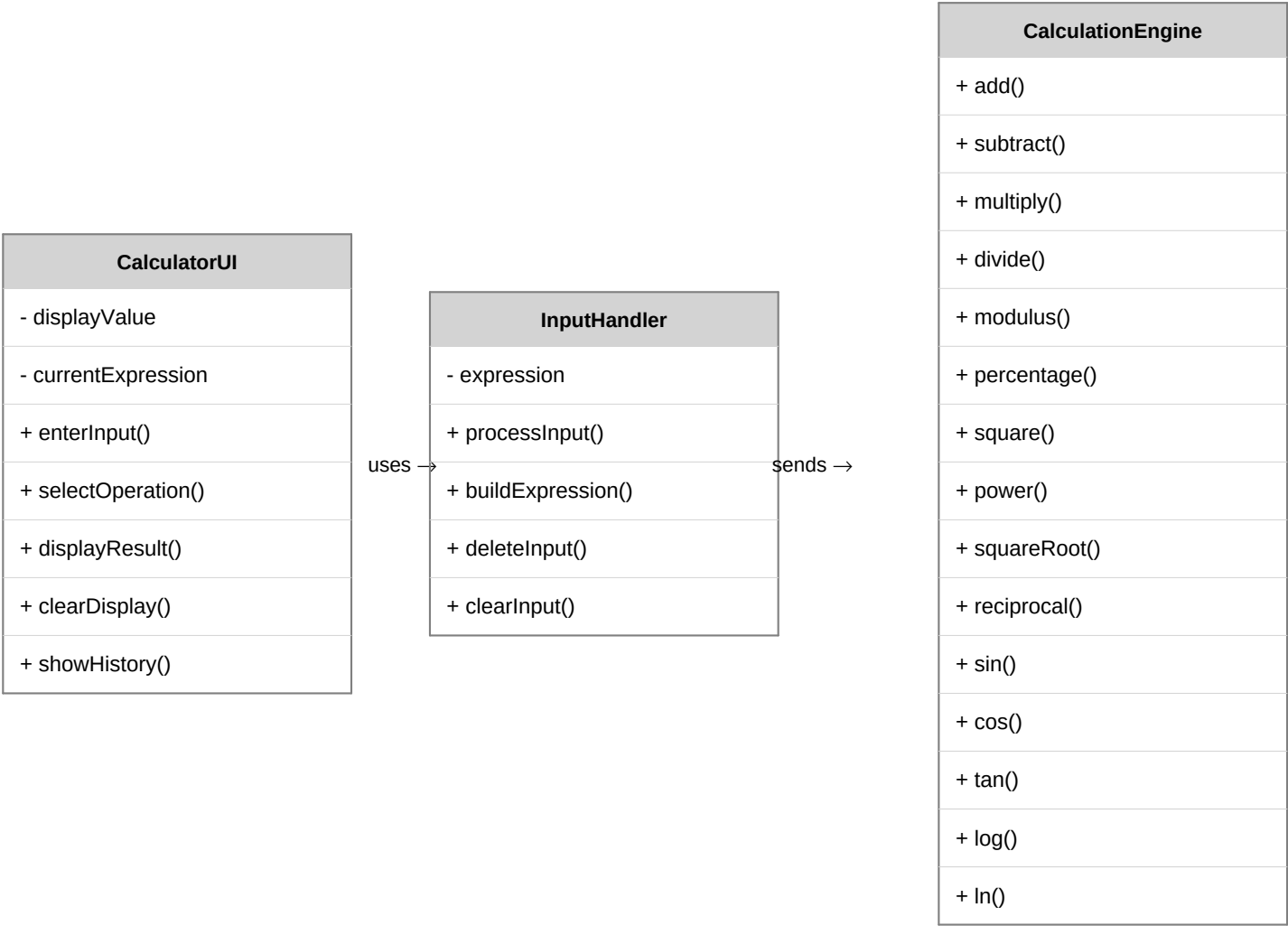
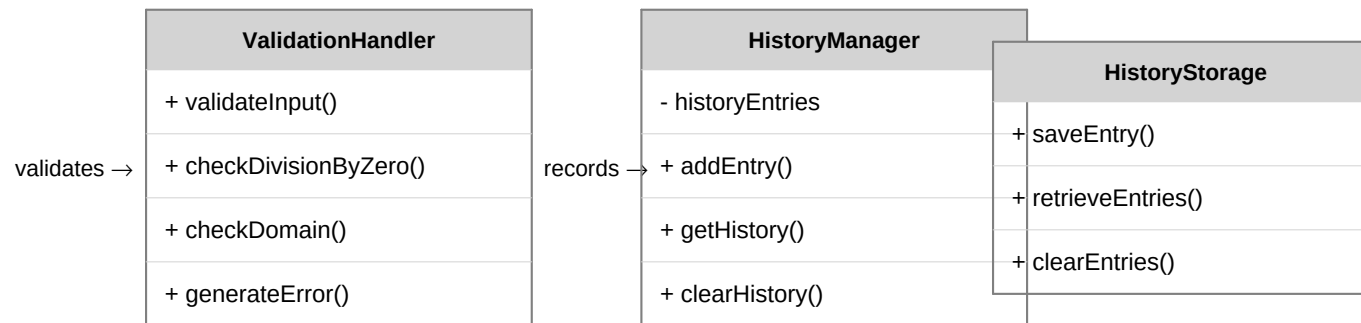


UML DIAGRAMS

Smart Scientific Calculator | Incremental Software Development Model | SEPM

1. Class Diagram





The class diagram separates presentation, input handling, calculation, validation, and history responsibilities. The calculation engine contains the mathematical operations introduced across the project increments.

2. Sequence Diagram — Normal Calculation

User	Calculator UI	Input Handler	Calculation Engine	Validation	History
Enter 25	receiveInput()	processInput()			
Select +	selectOperation()	buildExpression()			
Enter 15	receiveInput()	processInput()			
Press =	calculate()	sendExpression()	calculate(25+15)	validate()	
			result = 40	valid	addEntry(25+15=40)
	display 40				

Scenario: User performs $25 + 15 = 40$. The sequence demonstrates interaction between the user interface, input handler, calculation engine, validation, and history components.

3. Sequence Diagram — Invalid Calculation

User	UI	Input Handler	Calculation Engine	Validation
Enter 20 ÷ 0	receive input	build expression	calculate(20÷0)	
			validate operation	checkDivisionByZero()
				INVALID
	show error		return error	division by zero
See error	display "Cannot divide by zero"			

4. Activity Diagram — Calculator Operation



5. UML-to-Requirements Traceability

UML Element	Related Requirements / Use Cases
CalculatorUI	FR-16, FR-17, FR-20; UC-08 and all calculation use cases.
InputHandler	FR-01 to FR-18; UC-01 to UC-09.
CalculationEngine	FR-01 to FR-15; UC-01 to UC-07.
ValidationHandler	FR-19; UC-04, UC-06, UC-07, UC-10.
HistoryManager / HistoryStorage	FR-18; UC-09.
Normal Calculation Sequence	UC-01 to UC-07 — normal calculation interaction.
Invalid Calculation Sequence	UC-10 and FR-19 — invalid operation handling.
Activity Diagram	Overall calculator workflow from input through result and history.